

[Zn₁K₂]Fe₃(PO₄)₂(P₂O₇) as a Zn²⁺/K⁺ dual-ion cathode with enhanced diffusion kinetics and structural stability for aqueous zinc-ion batteries

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ABSTRACT

Aqueous zinc-ion batteries (AZIBs) have gained attention as safe and cost-effective alternatives to conventional lithium-ion batteries. However, their practical performance is often hindered by sluggish Zn²⁺ diffusion, primarily due to the large hydrated ionic radius and strong electrostatic interactions. To address this, we present a polyanion-based cathode material, [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇), that enables a reversible dual-ion intercalation mechanism involving both Zn²⁺ and K⁺. Compared to Zn²⁺, monovalent K⁺ interacts more weakly with the PO₄/P₂O₇ framework and desolvates more readily at the electrolyte–cathode interface, enabling the Zn²⁺/K⁺ dual-ion system to achieve faster kinetics than the Zn²⁺-only system. This leads to significantly enhanced power capability and structural reversibility under the AZIB system. The discharge capacity of [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇) is 100.4 mAh g^{−1} at C/5 (1C = 100 mA g^{−1}), corresponding to de/intercalation of 2 mol K⁺ and 0.5 mol Zn²⁺. Even at 5C, it maintains a capacity of 70 mAh g^{−1}, outperforming the Zn²⁺-only analogue Zn₂Fe₃(PO₄)₂(P₂O₇) (55 mAh g^{−1}). Galvanostatic intermittent titration technique measurements further confirm the enhanced ionic diffusion kinetics enabled by the Zn²⁺/K⁺ dual-ion system. Moreover, the capacity retention of [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇) is 84.6 % after 100 cycles, while Zn₂Fe₃(PO₄)₂(P₂O₇) deliver only 56.1 %. Advanced structural analyses reveal highly reversible lattice evolution during cycling, supporting the proposed dual-ion storage mechanism.

1. Introduction

Growing environmental concerns over fossil fuel consumption have spurred active research into lithium-ion batteries (LIBs) as eco-friendly energy storage systems (ESSs) [1–4]. However, as LIB applications extend from portable electronics to electric vehicles (EVs) and grid-scale ESSs, issues such as high cost and limited resource availability have become increasingly problematic due to the geographically concentrated and finite lithium supply [5–7]. Moreover, safety concerns arising from thermal runaway and explosion incidents have underscored the need for safer alternatives [8,9]. To address these issues, significant attention has been directed toward the development of aqueous batteries, which use water-based electrolytes instead of conventional

flammable organic liquid electrolytes [10,11].

Among various aqueous battery systems, aqueous zinc-ion batteries (AZIBs) have emerged as attractive candidates due to their affordability, safety, and sustainability [12–14]. Zinc (Zn) is an abundant and cost-stable resource, and it serves as a metal anode with high gravimetric and volumetric capacities (820 mAh g^{−1} and 5855 mAh cm^{−3}) [15, 16]. Furthermore, its intrinsic safety, environmental benignity, and suitable redox potential (−0.763 V vs. the standard hydrogen electrode, SHE) make it highly compatible with aqueous electrolytes [17–19]. Despite their merits, the practical application of AZIBs is hindered by several significant drawbacks, particularly the instability of cathode structures and their limited power capability [20]. One major challenge stems from the strong electrostatic attraction between divalent-Zn²⁺ and

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the cathode host material [21–23]. As divalent Zn^{2+} are intercalated into the cathode, this interaction can cause large structural damage or irreversible phase transitions [24–27]. Moreover, the high charge density of Zn^{2+} leads to a large and strongly bound hydration shell, making desolvation at the electrolyte–cathode interface energetically demanding and slowing interfacial ion transport [28–30].

Recent studies have demonstrated that incorporating monovalent alkali ions (K^+ , Na^+) into the electrolyte facilitates co-intercalation with Zn^{2+} , enhancing the electrochemical performance of AZIBs [31–35]. Specifically, incorporating K^+ into the cathode enhances bulk ion diffusion by increasing the interlayer distance and reducing electrostatic interactions with the host crystal structure, which lowers the migration barrier and improves reaction kinetics [31,34]. Compared to Zn^{2+} , monovalent K^+ has a lower charge density and a larger ionic radius, allowing it to interact more weakly with the lattice and occupy sites with lower binding energy [36–38]. This inherently facilitates faster solid-state ion transport and more reversible de/intercalation within the cathode. In addition to these bulk transport benefits, K^+ has a higher (less negative) hydration free energy (-321 kJ mol^{-1}) than Na^+ (-405 kJ mol^{-1}) and Li^+ (-515 kJ mol^{-1}), indicating weaker solvation in aqueous solutions [24,39]. The relatively small hydrated radius of K^+ in aqueous electrolyte facilitates easier desolvation at the electrolyte–cathode interface, further supporting rapid interfacial ion transport [40,41]. It suggests that incorporating K^+ into the intercalation mechanism can serve as an effective strategy to enhance both reaction kinetics and structural durability in AZIBs. Furthermore, the smaller hydration radius of K^+ compared to Na^+ (3.58 Å) and Zn^{2+} (4.04 Å) enables greater ionic mobility in aqueous electrolytes, thereby promoting more efficient ion diffusion [24,36,42]. Therefore, we hypothesized that a cathode capable of stably accommodating both Zn^{2+} and K^+ for diffusion and de/intercalation within its structure can deliver the outstanding electrochemical performances under the AZIB system.

Motivated by this concept, we develop a new polyanion-based cathode material, $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP), rationally designed to accommodate reversible $\text{Zn}^{2+}/\text{K}^+$ dual-ion intercalation within the structure. The PO_4 and P_2O_7 groups within ZKFP can provide the three-dimensional (3D) and spacious diffusion pathways for both Zn^{2+} and K^+ , leading to facile ionic diffusion during charge/discharge processes [18, 43,44]. In addition, the strong P–O covalent bonds and robust 3D framework of ZKFP allow reversible lattice breathing during dual-ion intercalation, ensuring long-term structural integrity [45,46]. Furthermore, ZKFP utilizes Fe and K sources, which are earth-abundant, offering notable cost advantages. Compared to other 3D frameworks such as Metal–Organic Frameworks (MOFs) and Prussian Blue Analogues (PBAs), ZKFP benefits from the strong covalent P–O bonds and robust phosphate/pyrophosphate framework, which provide high structural stability even under repeated Zn^{2+} intercalation. In addition, its simple solid-state synthesis using earth-abundant Fe and K sources offers superior cost-effectiveness, making it a more practical candidate for AZIB cathodes [47–50].

Using first-principles calculations and bond-valence energy landscape (BVLE) analysis, we demonstrated that ZKFP exhibits extensive three-dimensional diffusion pathways for both K^+ and Zn^{2+} , implying facile diffusion of these ions within the structure. Electrochemical measurements revealed that ZKFP delivered a specific capacity of approximately 100 mAh g^{-1} at a C/5 rate ($1\text{C} = 100 \text{ mAh g}^{-1}$) in the $\text{Zn}^{2+}/\text{K}^+$ dual-ion electrolyte, corresponding to de/intercalation of 2 mol K^+ and 0.5 mol Zn^{2+} . This electrochemical result was also supported by first-principles calculations, which revealed that K^+ deintercalation from ZKFP occurs prior to Zn^{2+} deintercalation and is thermodynamically more favorable. Even at a high rate of 5C, the capacity remained close to $\sim 70 \text{ mAh g}^{-1}$, which is 70 % of the capacity at C/5. In terms of cyclability, the discharge capacity of ZKFP was retained at 84.6 % of the initial capacity after 100 cycles.

In particular, comparison with Zn^{2+} -only analogue $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZFP) demonstrated that $\text{Zn}^{2+}/\text{K}^+$ dual-ion intercalation

mechanism under the AZIB system enhances both power capability and cycling performance of ZKFP. Moreover, ZKFP exhibited fully reversible structural changes in XRD during cycling, whereas ZFP showed irreversible peak shifts and internal strain accumulation, highlighting the enhanced structural reversibility achieved by K^+ co-intercalation. These findings confirm that the $\text{Zn}^{2+}/\text{K}^+$ dual-ion intercalation mechanism is essential for enhancing both electrochemical performance and structural stability under the AZIB system.

2. Material and methods

2.1. Synthesis of $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$

$\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (NFPP) was obtained by solid state method. $\text{Na}_4\text{P}_2\text{O}_7$ (Duksan Pure Chemicals, 98 %), $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (Sigma Aldrich, 99 %) and $\text{NH}_4\text{H}_2\text{PO}_4$ (Sigma Aldrich, 98.5 %) in a molar ratio of 1:3:2 were used as precursors, and 5 wt % of pyromellitic acid (PA) (Alfa Aesar, 96 %) were mixed by high-energy ball milling operating at 400 rpm for 12 h with silicon-nitride balls, and sealed in an Ar-filled glove box. The mixed powder was pelletized and heat treated in an Ar atmosphere at 300°C for 5 h to remove H_2O . After calcination, the powder was ground well with a mortar and 5 wt % of PA was added and re-mixed under same condition with first mixing method. The remixed powder was pelletized and reheated to 550°C for 12 h under an Ar flow. To replace Na^+ ions in NFPP with K^+ ions and/or Zn^{2+} , an electrochemical ion exchange process was conducted using aqueous zinc-ion cells. NFPP electrodes were assembled in a Zn metal | aqueous electrolyte | NFPP configuration and cycled for 5 charge/discharge cycles at 100 mA g^{-1} in a voltage range of 0.4–1.9 V. For ZKFP, a Zn/K dual-ion electrolyte (0.5 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2 + 18 \text{ M K}(\text{CF}_3\text{SO}_3)$) was used, resulting in simultaneous Na^+ removal and $\text{Zn}^{2+}/\text{K}^+$ co-intercalation, forming $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. For ZFP, a Zn-only electrolyte (0.5 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$) was used, leading to Na^+ extraction and Zn^{2+} intercalation, forming $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

2.2. Materials characterization

The elemental composition of substances like Na, K, Zn, Fe, P, and O was analyzed using an inductively coupled plasma atomic emission spectrometer (ICP-AES; OPTIMA 8300, Perkin-Elmer). The size and morphology of the materials were performed using a FE-SEM (Gemini SEM 560, ZEISS) at 2 keV, conducted at the National Center for Inter-University Research Facilities (NCIRF) affiliated with Seoul National University. HR-TEM image and FFT were measured by 200 kV STEM (JEM-ARM300F, JEOL) at the Advanced Facility Center for Quantum Technology at the Sungkyunkwan University. Prior to the measurements, the samples were dispersed in ethanol with the aid of an ultrasonic device. Following this, a droplet of the suspension was applied onto a copper TEM grid coated with carbon, and the grid was left to air-dry at room temperature overnight to facilitate the evaporation of ethanol. The crystal structure of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were examined through X-ray diffraction (XRD) using a PANalytical instrument with $\text{Mo K}\alpha$ radiation (wavelength = $0.70932 \text{ \AA}$). The data were collected in the 2θ range of 4.61° to 34.32° with a 0.01° step size. Subsequently, the XRD patterns were reevaluated by converting the angles to $\text{Cu K}\alpha$ radiation ($\lambda = 1.54178 \text{ \AA}$) for comparison with prior research. The XRD refinement data were analyzed using the Fullprof Rietveld program. The samples were prepared in electrode form and analyzed by X-ray photoelectron spectroscopy (XPS) using an Axis Supra (Kratos, UK) at the National Center for Inter-University Research Facilities (NCIRF), Seoul National University.

2.3. Electrochemical characterization

The cathode was mixed with Active materials, carbon nanotubes

(CNTs), polyvinylidene fluoride (PVdF) in a mass ratio of 75:15:10, and N-methyl-2-pyrrolidone (NMP) was used as a solvent. The slurry was coated onto Ti foil (Welcos, Korea) and dried under vacuum conditions at 100 °C overnight. The zinc foil (MTI Korea) was used as an anode. The dried electrodes were cut into disks: a diameter of 10 ϕ (mm) for the cathode and 14 ϕ (mm) for the zinc foil anode. All electrochemical measurements were conducted using cathodes with an active material loading of approximately 2 mg cm⁻², which is within the typical range (1–3 mg cm⁻²) reported in previous studies [51,52]. The CR2032 coin-type cells were assembled by Zinc foil, ZKFP cathode, glass fiber (Whatman GF/F), and an aqueous electrolyte. Galvanostatic charge/discharge experiments were performed across a range of current densities, from 20 to 500 mA g⁻¹, within the voltage range of 0.4–1.9 V (vs. Zn²⁺/Zn). Throughout the tests, the charging current density was maintained at a steady 20 mA g⁻¹. Battery cycling performance was evaluated using an automated charge/discharge testing system (WBCS

3000, WonATech).

Galvanostatic intermittent titration technique (GITT), introduced by Weppner and Huggins, was employed to evaluate the diffusion kinetics of Zn²⁺ and K⁺ ions in ZKFP. Constant current pulses (20 mA g⁻¹, 10 min) were applied with a relaxation time of 30 min at each step. *Ex-situ* X-ray absorption spectroscopy (XAS) measurements were carried out at the Fe K-edge to investigate the oxidation state and local coordination environment of Fe in [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇). The experiments were conducted at the 10C beamline of the Pohang Accelerator Laboratory (PAL), South Korea, using Fe metal foil as a reference. The XAS spectra were collected in fluorescence mode under a storage ring energy of 2.5 GeV and a beam current of 200 mA, covering both the X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) regions. The data acquired from XAFS measurements were subsequently analyzed using the Athena software package. It is important to note that the samples were prepared in the form of electrodes.

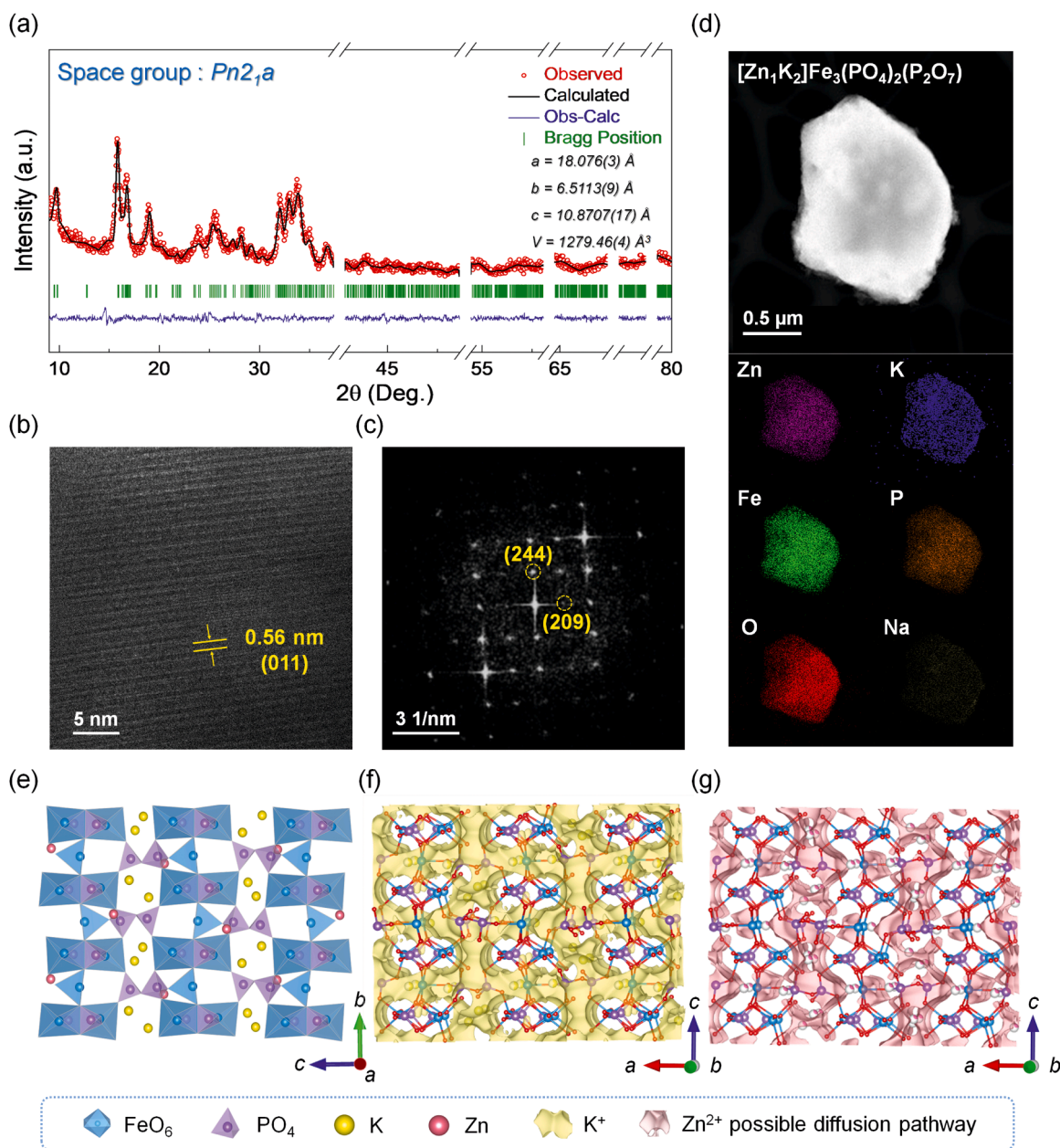


Fig. 1. Characterization of [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇): a) XRD pattern and Rietveld refinement results b) high-resolution TEM (HRTEM) images and c) the associated fast Fourier transform (FFT). d) TEM-EDS elemental mapping images. e) Crystal structure. Three-dimensional BVEL analyses with all possible f) K⁺ and g) Zn²⁺ diffusion pathways in the crystal structure.

2.4. Computational details

The density functional theory (DFT) calculations were conducted with the projector augmented wave (PAW) method, which is implemented in the Vienna Ab initio Simulation Package (VASP). The supercell shape of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was $1 \times 2 \times 1$, and the k-point grid was automatically set to be $6 \times 3 \times 3$ by VASP. While the DFT calculations, the total alkali ion content of $\text{Zn}_x\text{K}_y\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ was changed with a 0.5 mol step, and the calculations were being conducted with a maximum of 20 configurations that have the lowest electrostatic energies in each composition. Those configurations were selected through the cluster expansion using CASM software and comparing Ewald summation. We visualized the localization of the *d*-orbital in Fe ions with the effective U values of 4.3 eV. The kinetic energy cut-off was set as 500 eV values to characterize the interaction between the ionic cores and the valence e^- , and the ionic relaxation loops were stopped

$$V = -\frac{E[\text{Zn}_{x_2}\text{K}_{y_2}\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)] - E[\text{Zn}_{x_1}\text{K}_{y_1}\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)] - (x_2 - x_1)E[\text{Zn}] - (y_2 - y_1)E[\text{K}]}{(x_2 - x_1)F} \quad (1)$$

when the norms of all the forces were smaller than 0.03 eV.

3. Result and discussion

3.1. Crystal structure and morphology of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP)

To prepare $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP), the electrochemical ion-exchange process under the $\text{Zn}^{2+}/\text{K}^+$ dual-ion electrolyte was performed, since the direct synthesis of ZKFP using the typical solid-state synthesis methods results in impurity phases rather than the desired ZKFP phase (Figure S1). The detailed preparation process of ZKFP is described in the Experimental section.

As shown in Fig. 1a, it was confirmed that ZKFP exhibited the orthorhombic phase with space group $\text{Pn}2_1\text{a}$, which is similar to the $\text{Na}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (NFPP) (Figure S2). The elemental ratio of Zn:K:Na:Fe in ZKFP was estimated to be approximately 1:2:0:3 by inductively coupled plasma (ICP) spectroscopy analysis, indicating successful preparation of ZKFP (Table S1). The structural characteristics of ZKFP were examined using Rietveld refinement. The refined lattice parameters were determined as $a = 18.076(3) \text{ \AA}$, $b = 6.5113(9) \text{ \AA}$, $c = 10.8707(17) \text{ \AA}$, corresponding to a unit cell volume of $1279.46(4) \text{ \AA}^3$. Detailed atomic coordinates, thermal parameters, and site occupancies for the crystal structure of ZKFP are presented in Supporting Table S2. The low R-factors ($R_p = 10.1 \%$, $R_i = 7.99 \%$, $R_f = 2.19 \%$ and $\chi^2 = 2.07 \%$) reflect the high reliability and accuracy of the Rietveld refinement.

In the high-resolution transmission electron microscopy (HR-TEM) image (Fig. 1b), lattice fringes with an interplanar spacing of approximately 0.56 nm are observed, corresponding to the (011) plane of the ZKFP structure. Additional analysis using fast Fourier transform (FFT) of the crystal lattice (Fig. 1c) indicates that the (244) and (209) planes correspond to a typical NASICON structure, which aligns with the results obtained from Rietveld refinement. Fig. 1d clearly shows the smooth surface of the flakes, and elemental mapping reveals the uniform distribution of Zn, K, P, Fe, and O in the ZKFP particles. In addition, scanning electron microscopy (SEM) analysis revealed that the particle size (D50) of ZKFP was approximately 1.18 μm (Figure S3), consistent with the TEM results. Using TEM-based energy-dispersive X-ray spectroscopy (TEM-EDS), it was also verified that the elemental ratio of Zn:K:Fe in ZKFP was found to be approximately 1:2:3, which is consistent with the results obtained from the ICP results.

As shown in Fig. 1e, the crystal structure of ZKFP consists of $\text{Fe}_3\text{P}_2\text{O}_{13}$

layers aligned along the *bc* plane. Within these $\text{Fe}_3\text{P}_2\text{O}_{13}$ layers, FeO_6 octahedra are connected to P_2O_7 diphosphate units along the *a*-axis. To explore the diffusion behavior of K^+ and Zn^{2+} ions within the crystal structure, we performed bond valence energy landscape (BVEL) analyses. The BVEL analysis results reveal that the structure contains sufficiently large vacant spaces and three-dimensional pathways for both K^+ (Fig. 1f) and Zn^{2+} (Fig. 1g) diffusion. This suggests the facile de/intercalation of K^+ and Zn^{2+} ions in ZKFP during charge/discharge.

3.2. $\text{Zn}^{2+}/\text{K}^+$ storage dual-ion storage mechanism and power capability of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP)

To verify the theoretical electrochemical properties of ZKFP, we performed first-principles calculations. We calculated the theoretical redox potentials of ZKFP during Zn^{2+} and K^+ de/intercalation, employing the following Eq. (1):

Here, V stands for the average redox potential within the range of $(x_1 + y_1) \leq z \leq (x_2 + y_2)$, while E denotes the formation energy for the most stable configurations of each ZKFP composition. Additionally, $E[\text{Zn}]$ and $E[\text{K}]$ represent the energies of Zn and K metal, and F means the Faraday constant.

Fig. 2a illustrates the convex hull plot, showing the changes in formation energy as K^+ ions are initially extracted from the ZKFP structure, followed by Zn^{2+} extraction. The x-axis of the plot represents $(x + y)$ in $\text{Zn}_x\text{K}_y\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, where $x + y = 3$ mol, corresponding to the initial total amount of mobile ions in the structure. In this case, when K^+ ions are preferentially deintercalated first, 0.5 mol of Zn^{2+} must remain in the structure to ensure its stability. Since 1 mol of electrons must remain in the structure, no further extraction is possible [53]. Fig. 2b shows the convex hull plot for the case where Zn^{2+} ions are initially deintercalated, followed by K^+ ions. In contrast to K^+ -first deintercalation, when Zn^{2+} ions are deintercalated first, 1 mol of K^+ must remain in the structure to ensure its stability. Fig. 2c displays the theoretical redox potential corresponding to each deintercalation pathway. In the case where K^+ ions are initially deintercalated, 2 mol of K^+ ions are first extracted, followed by the removal of 0.5 mol of Zn^{2+} ions, resulting in a higher redox potential of 1.863 V.

In the Zn-first deintercalation case, 1 mol of Zn^{2+} ions is initially extracted, followed by the removal of 1 mol of K^+ ions, leading to a lower redox potential of 1.339 V. Notably, the K^+ -first extraction enables both Zn^{2+} and K^+ ions to be reversibly involved within the voltage window of 0.4–1.9 V (vs. Zn^{2+}/Zn), resulting in a higher average voltage compared to the Zn^{2+} -first extraction case. This sequence confirms that K^+ ions preferentially are deintercalated before Zn^{2+} ions, thereby facilitating more favorable ionic transport during charge/discharge cycles. This demonstrates that prioritizing K^+ deintercalation is advantageous for maximizing the electrochemical utilization of both cations and achieving superior battery performance.

As shown in Fig. 2d, the theoretical redox potential of ZKFP based on the deintercalation of K^+ prior to Zn^{2+} is in good agreement with the experimentally measured galvanostatic intermittent titration technique (GITT) data. This indicates that 2 mol of K^+ ions are first deintercalated from the structure, followed by 0.5 mol of Zn^{2+} ions. The voltage plateaus corresponding to each ion are also marked in Fig. 2d using shaded regions, reinforcing the sequential nature of K^+ and Zn^{2+} deintercalation.

Moreover, using the nudged elastic band (NEB) method based on

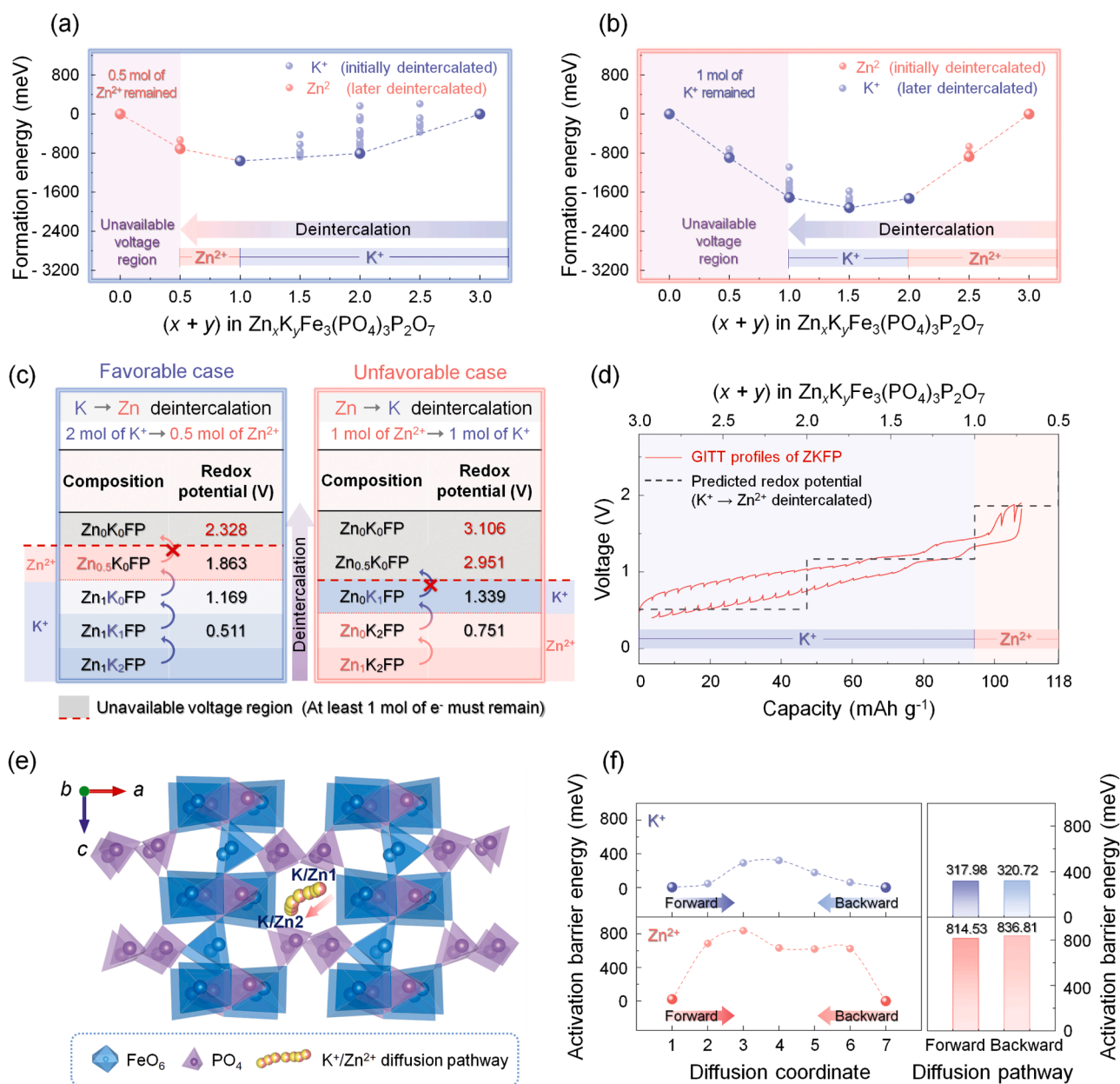


Fig. 2. Convex hull diagrams showing the formation energy changes when a) K⁺ and b) Zn²⁺ are preferentially deintercalated from the [Zn₁K₂]Fe₃(PO₄)₂(P₂O₇) (ZKFP) structure. c) Theoretical redox potential variations corresponding to the deintercalation pathways shown in a) and b). d) Comparison between the calculated redox potentials and the experimentally measured GITT profile, confirming the sequential deintercalation of K⁺ followed by Zn²⁺. e) Predicted diffusion pathways of K⁺ and Zn²⁺ ions in the ZKFP structure. f) Calculated activation energy barriers for K⁺ and Zn²⁺ diffusion in ZKFP.

first-principles calculations, we confirmed that K⁺ diffusion in the ZKFP structure is more favorable than Zn²⁺ diffusion. The diffusion pathways of K⁺ and Zn²⁺ within the orthorhombic ZKFP lattice are illustrated in Fig. 2e, and their corresponding activation energy barriers are presented in Fig. 2f. The calculated energy barrier for K⁺ diffusion along the K1–K2 pathway was approximately 320.72 meV, whereas that for Zn²⁺ diffusion was significantly higher, at around 836.81 meV, nearly 2.6 times greater than that of K⁺. These NEB results indicate that although Zn²⁺ diffusion is thermodynamically feasible and structurally stable within the host lattice, its higher migration barrier leads to slower ion transport kinetics. This finding suggests that the concurrent utilization of Zn²⁺ and K⁺ ions, rather than Zn²⁺ alone, can synergistically enhance the power capability of AZIBs.

To investigate the power capability of ZKFP, the specific capacities were measured at various current rates ranging from C/5 to 5C (1C =

100 mA g⁻¹) (Fig. 3a and Figure S4). The initial discharge capacity of ZKFP reached 100.4 mAh g⁻¹ at C/5 with a coulombic efficiency of 97.5 %. To assess the competitiveness of this result, we compared the initial capacities of ZKFP with those of other representative polyanion-based cathodes reported for aqueous zinc-ion batteries. As shown in Table S3, ZKFP exhibits one of the highest initial capacities among reported polyanion-based AZIB cathodes. Even at a high rate of 5C, ZKFP delivered a specific capacity of approximately 70 mAh g⁻¹, which corresponds to about 70 % of the capacity achieved at C/5. To compare the high-rate performance of ZKFP with other polyanion-based cathodes reported for aqueous zinc-ion batteries, we summarized representative literature results in Table S4. Compared to prior works, ZKFP exhibits one of the highest discharge capacities at high current densities and superior rate retention, highlighting the advantages of the Zn²⁺/K⁺ dual-ion intercalation strategy. Furthermore, electrode-level optimization with a

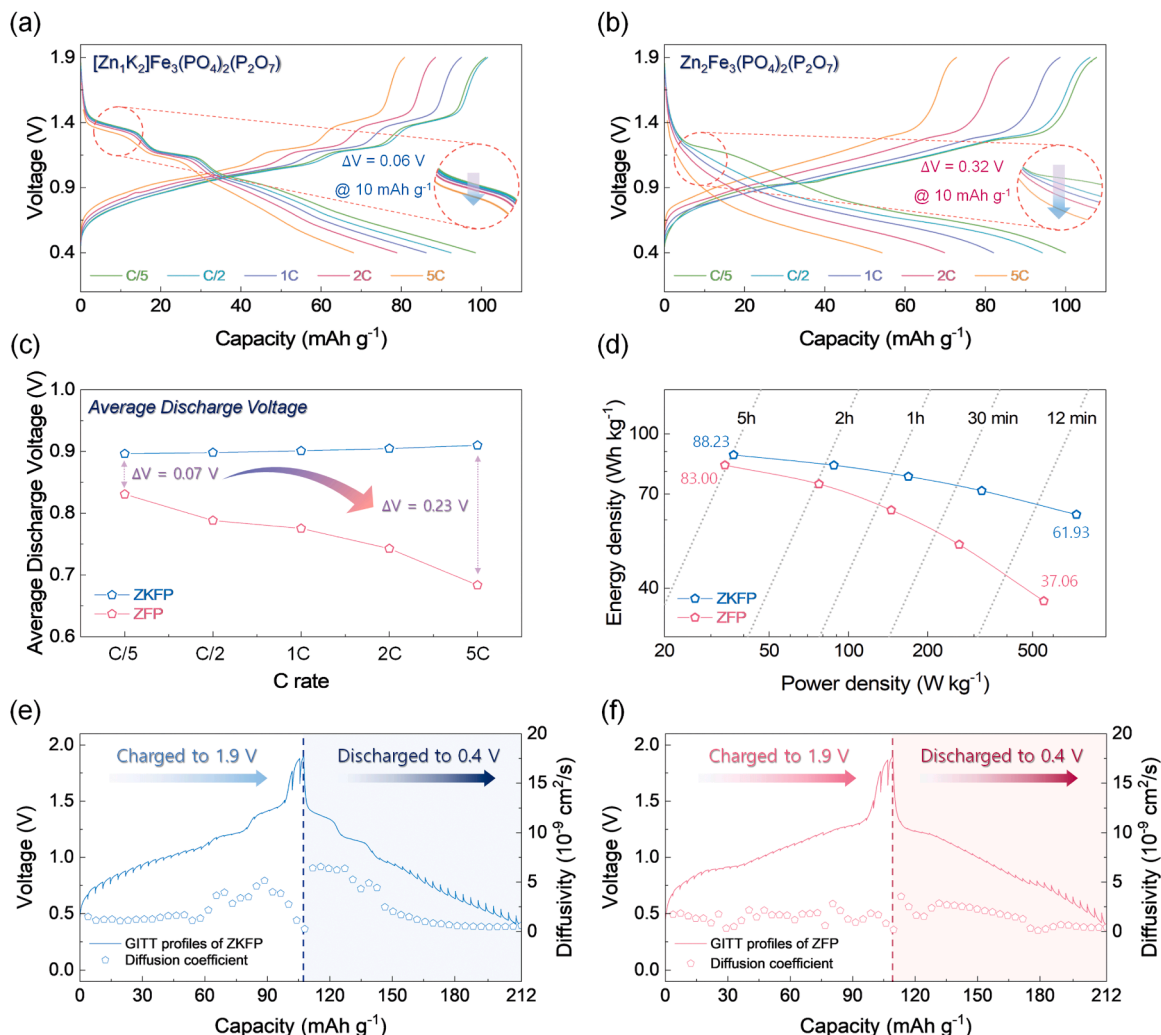


Fig. 3. Electrochemical performance of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP) and $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZFP): a) Charge/discharge profiles of ZKFP at various current densities (C/5 to 5C). b) Charge/discharge profiles of ZFP under the same C-rate conditions. c) Average discharge voltages of ZKFP and ZFP at various C-rates. d) Energy density of ZKFP and ZFP as a function of discharge C-rate. GITT profiles at C/5 with corresponding diffusion coefficients of e) ZKFP and f) ZFP, respectively.

hybrid conductive additive composed of CNTs and Super P (5:5 wt ratio) also enhanced rate capability, enabling more efficient charge transport and better capacity retention at high current densities (Figure S5). This improvement is attributed to the synergistic effects of CNTs, which provides efficient electron pathways, and Super P, which enhances contact uniformity and electrode cohesion [54]. The redox stability of ZKFP was further confirmed by cyclic voltammetry at a scan rate of 0.3 mV s⁻¹ over 10 cycles. As shown in Figure S6, the CV curves remained highly overlapped with negligible peak shifts, demonstrating excellent electrochemical reversibility and structural stability during prolonged cycling under the AZIB system.

Particularly, we demonstrated that the power capability of ZKFP under the AZIB system is more outstanding than that of only Zn-based $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZFP). ZFP was prepared through the electrochemical ion-exchange process like ZKFP, and the detailed preparation process is described in Experimental section. It was verified that ZFP has consistent crystal structure with ZKFP (Figure S7). As shown in Fig. 3b and Figure S8, ZFP just delivered the capacity of 55 mAh g⁻¹ at 5C, corresponding to ~55 % of its capacity at C/5. Moreover, the discharge voltage gap (ΔV) between the C/5 and 5C charge/discharge profiles of ZKFP was found to be significantly smaller than that of the Zn²⁺-only ZFP. Fig. 3c-d shows the average discharge voltages and energy densities of ZKFP and ZFP as a function of the current rate. It was verified that the difference in average discharge voltages and energy densities between

ZKFP and ZFP increases progressively with higher discharge current rate.

Furthermore, difference of the ionic diffusion kinetics between ZKFP and ZFP were investigated through the GITT analyses. The apparent diffusion coefficient (D) was calculated using Fick's second law according to the following equation:

$$D = \frac{4}{\pi\tau} \left(\frac{mV}{MS} \right)^2 \left(\frac{\Delta E_s}{\Delta E_t} \right)^2, \quad \left(t \ll \frac{L^2}{D} \right) \quad (2)$$

In the Eq. (2), D represents the diffusion coefficient of $\text{Zn}^{2+}/\text{K}^+$, τ represents the charging/discharging duration (s), m represents the amount of active material on the electrode (g), V represents the molar volume (cm³ mol⁻¹) of the electrode, M corresponds to the molecular weight (g mol⁻¹) of electrode, S represents the surface area of contact between the electrode and electrolyte, ΔE_s represents the voltage change during a single step, and ΔE_t reflects the cumulative voltage change under applied constant current conditions. In this study, the electrode area (S) was defined as the geometric area of the cathode disk (πr^2), where $r = 0.5$ cm for a 10 mm diameter electrode. Accordingly, the calculated diffusion coefficients represent apparent values based on the full geometric electrode area rather than intrinsic electrochemically active surface area.

The GITT profiles show that the ZKFP cathode exhibits diffusion coefficients in the range of 10⁻¹⁰ to 10⁻⁹ cm² s⁻¹ across the entire voltage

range (Fig. 3e). In contrast, the Zn^{2+} -only ZFP delivers lower diffusion coefficients (10^{-11} to low $10^{-10} \text{ cm}^2 \text{ s}^{-1}$) throughout the voltage window (Fig. 3f), indicating sluggish ionic transport. These results demonstrate that the $\text{Zn}^{2+}/\text{K}^{+}$ dual-ion system in ZKFP significantly enhances ionic diffusion kinetics and power capability under the AZIB system.

The improved diffusivity of K^{+} is attributed to its single positive charge (+1), which leads to weaker electrostatic interactions with the host lattice compared to the doubly charged Zn^{2+} (+2) [55]. In contrast, Zn^{2+} ions experience stronger electrostatic binding to the lattice oxygen atoms, which hinders their migration through the framework and thereby limits diffusion kinetics [56–58].

3.3. Cyclability and structural stability of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP)

The cyclability of the ZKFP cathode was assessed at C/5, as shown in Fig. 4a. After 100 cycles, the ZKFP delivered the capacity retention of 84.6 % with a high coulombic efficiency above 99 %, indicating stable cycle-performance under the AZIB system. On the other hand, in the case of ZFP, its capacity retention was only 56.1 % of its initial capacity under the same conditions. The difference in long-term electrochemical stability and reversibility between ZKFP and ZFP became distinctly evident within 100 cycles, enabling a reliable comparison of their cycling behaviors. Moreover, their main difference between ZKFP and ZFP during prolonged cycling is the degree of voltage-decay. As shown in Fig. 4b, the average discharge voltage of ZKFP decreased marginally from 0.90 V to 0.85 V over 100 cycles, whereas ZFP exhibited a more pronounced

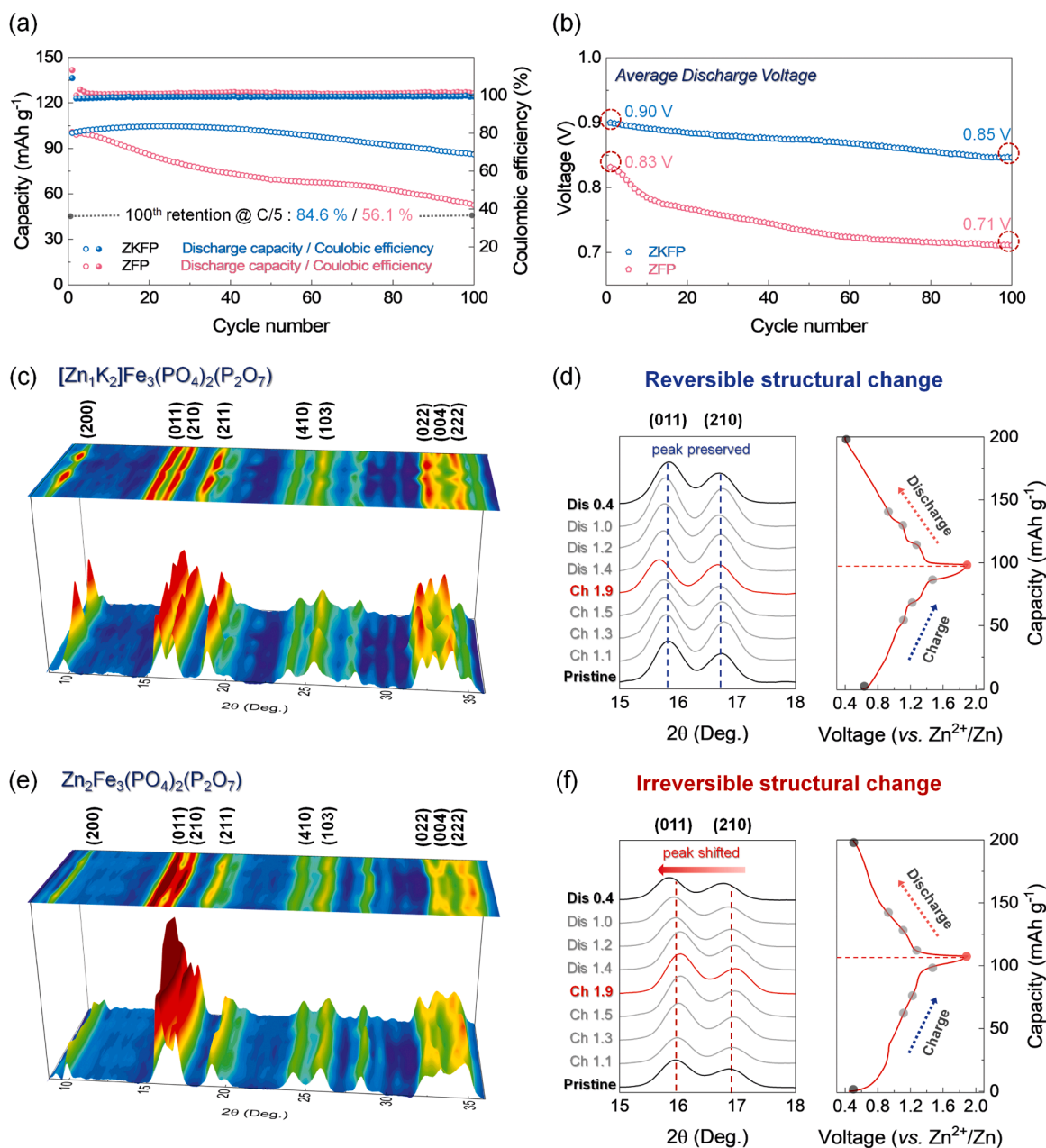


Fig. 4. a) Cycling performance of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP) and $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZFP) at C/5. b) Comparison of average discharge voltages. c) 3D contour maps of *ex-situ* XRD patterns of ZKFP collected at various charge/discharge states. d) Enlarged *ex-situ* XRD patterns of ZKFP showing reversible shifts in the (011) and (210) reflections during cycling at 0.2 C. e) 3D contour maps of *ex-situ* XRD patterns of ZFP collected at various charge/discharge states. f) Enlarged XRD patterns of ZFP showing irreversible shifts and structural distortion in (011) and (210) reflections.

drop from 0.83 V to 0.71 V. To further understand the origin of this performance difference, *ex-situ* SEM analysis was conducted after 100 cycles (Figure S9). The ZFP electrode displayed extensive surface cracking and particle degradation, suggesting significant mechanical stress accumulation and structural instability caused by repeated Zn^{2+} intercalation. Conversely, ZKFP maintained a more intact morphology with minimal cracking, indicating that K^+ co-intercalation helps alleviate internal strain and enhances mechanical resilience during cycling. Under the optimized electrode configuration using the hybrid conductive additive, cycling stability was further improved (Figure S10).

Ex-situ XRD measurements were performed to investigate the structural evolution and energy storage mechanism of ZKFP during charge/discharge. ZKFP electrodes were harvested at various charge/discharge states under a current density of C/5. As shown in Fig. 4c, XRD peaks corresponding to the (200), (011), (210), (211), (022), (004), and (222) planes are clearly observed in the pristine state. During cycling, the peak positions and intensities continuously evolve, indicating dynamic structural changes associated with the reversible de/intercalation of Zn^{2+} and K^+ ions (Fig. 4d). Notably, the (200) reflection exhibits reversible intensity change. It intensifies during the initial charging process, decreases at the fully charged state (1.9 V), and shows a similar intensity variation upon discharge. These cyclic intensity changes reflect a reversible lattice breathing behavior, suggesting dynamic atomic rearrangements along the (200) plane during dual-ion intercalation. In addition, the (011) and (210) peaks shift toward lower angles during charging and recovered to their original positions after discharge, indicating reversible structural change of ZKFP during charge/discharge.

To gain deeper insight into these structural variations, Rietveld refinement was applied to the *ex-situ* XRD data (Figure S11). Compared to the pristine state, the *a*-lattice, *b*-lattice, and *c*-lattice parameters show minimal changes of 0.20 %, 0.02 %, and 0.14 %, respectively, after discharge. The overall unit cell volume shows a difference of just 0.32 %,

indicating that the structural changes after charge and discharge are relatively small. These anisotropic lattice changes are attributed to the repulsive forces resulting from the deintercalation of Zn^{2+} (0.74 Å) and K^+ (1.33 Å), which reduce charge screening and increase interatomic distances. During discharge, all lattice parameters return close to their original values, confirming a highly reversible lattice breathing process. Moreover, it was observed that ZKFP exhibited only minor changes in its XRD pattern even after 100 cycles (Figure S12). The reversible peak shifts and volume changes observed in ZKFP during charge/discharge highlight its ability to accommodate internal stress and maintain structural stability during cycling, thereby contributing to its outstanding cycling performance and reversible electrochemical behavior.

In contrast to ZKFP, the ZFP structure showed minimal peak shifts and intensity changes throughout cycling (Fig. 4e-f). The (200) reflection remained weak and nearly unchanged, indicating limited structural flexibility along this plane. Although the (011), (210), (211), (022), and (222) peaks exhibited some shifts, these were less pronounced and irreversible. Notably, the (011) and (210) peaks failed to return to their original positions after discharge, suggesting incomplete structural recovery and possible residual strain or phase distortion. This restricted structural adaptability can be attributed to the sluggish Zn^{2+} diffusion and the absence of K^+ -assisted lattice dynamics. These comparative results clearly demonstrate the critical role of K^+ co-intercalation in enabling reversible lattice modulation and suppressing irreversible distortions. Regarding the lattice parameter changes (Figure S13), the ZFP structure shows more significant changes compared to the pristine state, with the *a*-lattice, *b*-lattice, and *c*-lattice parameters increasing by 0.16 %, 0.72 %, and 0.47 %, respectively, after cycling. The overall unit cell volume shows a difference of 1.36 %, indicating greater structural deformation and a larger volume change in ZFP during charge/discharge compared to ZKFP.

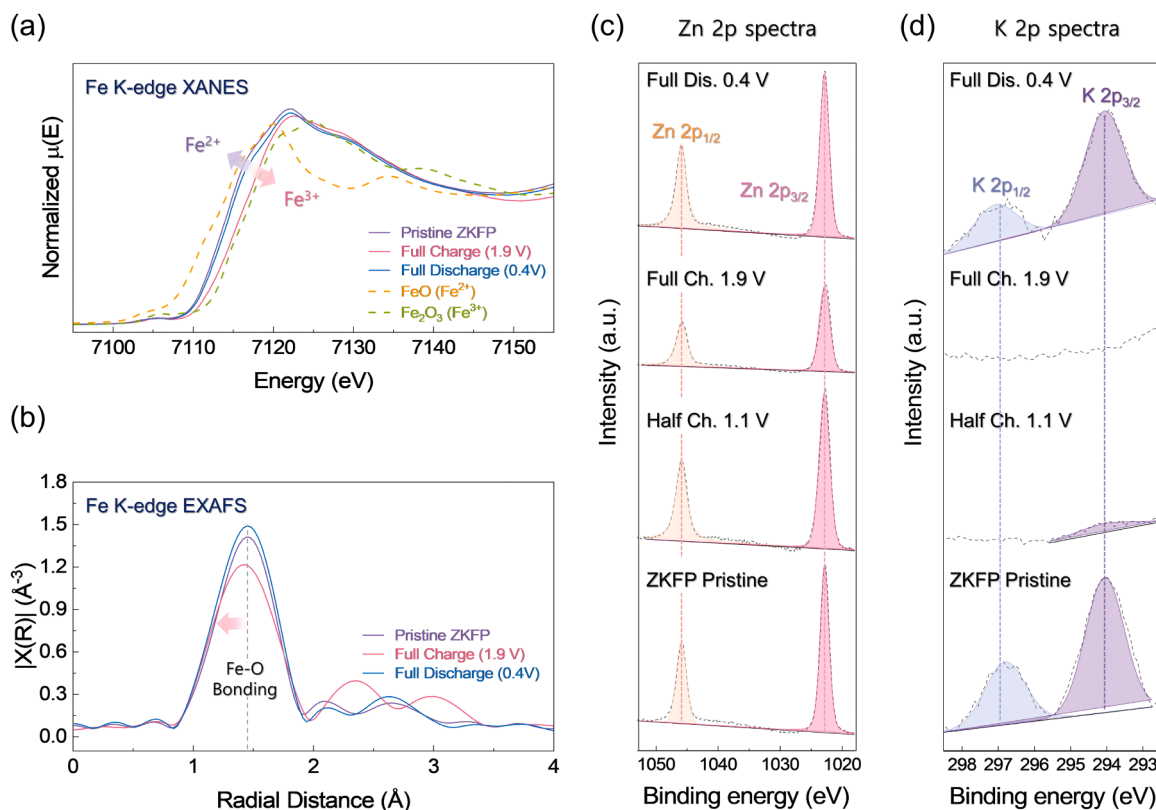


Fig. 5. a) *Ex-situ* Fe K-edge XANES spectra of various states of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP). b) Fe K-edge EXAFS analyses of various charge/discharge state of ZKFP. c) Zn 2p, and d) K 2p X-ray photoelectron spectroscopy (XPS) spectra.

3.4. Redox and intercalation mechanisms of $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP)

An *ex-situ* synchrotron-based XANES analysis was conducted to investigate the changes in the valence state of Fe ions in ZKFP during Zn^{2+} and K^+ de/intercalation. As shown in the Fe K-edge XANES spectra (Fig. 5a), the Fe K-edge shifted toward a higher energy level upon charging to 1.9 V and returned to its original position upon discharging to 0.4 V. These results indicate that a reversible $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox reaction occurs in ZKFP during Zn^{2+} and K^+ de/intercalation. The $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox reaction in ZKFP is further supported by first-principles calculations. As shown in Figure S14, we compared the integrated spin moments of Fe ions in $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, $[\text{Zn}_1\text{K}_0]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$, and $[\text{Zn}_{0.5}\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. Considering that Fe^{2+} and Fe^{3+} possess 6 and 5 electrons in the 3d orbital, respectively, the total spin moment of Fe ions in ZKFP increases upon Zn^{2+} and K^+ deintercalation, confirming the occurrence of the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox reaction.

Especially, the *ex-situ* Fe K-edge EXAFS results revealed that ZKFP undergoes reversible local structural changes during charge and discharge processes, indicating enhanced structural flexibility. As shown in Fig. 5b, the Fe–O bond distance shortened during charging, corresponding to the oxidation of Fe^{2+} to Fe^{3+} , and returned to its original value after discharging. This reversible modulation of the local Fe coordination environment is closely associated with the smooth redox reaction of Fe ions and contributes to the superior electrochemical stability of ZKFP.

Moreover, *ex-situ* X-ray photoelectron spectroscopy (XPS) analyses provide more detailed reaction mechanism on ZKFP. To experimentally verify the sequence of Zn^{2+} and K^+ de/intercalation during charge/discharge using *ex-situ* XPS analyses, we prepared the pristine, 1.1V-charged, 1.9V-charged and 0.4V-discharged ZKFP samples. As shown in Fig. 5c, the peaks at 1022.8 eV and 1045.9 eV correspond to the $\text{Zn}^{2+} 2p_{3/2}$ and $2p_{1/2}$ electrons, respectively [59]. Upon charging to 1.9 V (vs. Zn^{2+}/Zn), the peak intensities slightly decrease, while a recovery in intensity is observed after discharging to 0.4 V. The presence of Zn signals even at the fully charged state indicates that a fraction of Zn^{2+} ions remains embedded within the host structure. This suggests that the presence of residual Zn^{2+} ions is essential for maintaining the structural stability of ZKFP during charge. In particular, the K 2p XPS peak intensity was significantly reduced at the 1.1 V-charged (half-charged) state, whereas the Zn 2p XPS spectra showed negligible change at the same stage (Fig. 5d). The K 2p peaks at 296.9 eV and 294.1 eV correspond to K $2p_{3/2}$ and $2p_{1/2}$ electrons, respectively [60]. These results indicate that during charge, 2 mol K^+ ions are initially deintercalated from ZKFP, followed by the deintercalation of 0.5 mol Zn^{2+} ions, which is in good agreement with the first-principles calculation results.

4. Conclusion

In this study, we developed a new polyanion-based cathode material, $[\text{Zn}_1\text{K}_2]\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZKFP), for aqueous zinc-ion batteries (AZIBs) and systematically investigated its electrochemical and structural behavior in comparison with the Zn^{2+} -only analogue, $\text{Zn}_2\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ (ZFP). The $\text{Zn}^{2+}/\text{K}^+$ dual-ion system enhanced ion transport both within the bulk cathode and at the electrolyte–cathode interface, leading to faster diffusion kinetics and higher rate capability. First-principles calculations revealed that K^+ incorporation lowers migration barriers, accounting for the improved electrochemical performance, and also provided insights into the deintercalation mechanisms of both Zn^{2+} and K^+ , contributing to a deeper understanding of the dual-ion storage process.

In terms of the electrochemical properties, ZKFP delivered a discharge capacity of approximately 100 mAh g^{-1} at C/5 and retained 84.6 % of its capacity after 100 cycles, whereas ZFP showed only 56.1 % retention under identical conditions. Even at a high rate of 5C, ZKFP maintained around 70 % of its initial capacity, outperforming ZFP,

which retained only about 55 %. The voltage gap (ΔV) between C/5 and 5C charge/discharge profiles was significantly narrower for ZKFP (0.06 V) than for ZFP (0.32 V), indicating reduced polarization and faster reaction kinetics. The redox potential could be tuned through transition metal substitution, for example by partially replacing Fe with Mn or Co, in a manner similar to olivine-type $\text{Li}[\text{TM}]\text{PO}_4$ (TM = Fe, Mn, Co, etc.) [61], where variations in the TM species are known to shift redox potentials via their distinct electronic structures and inductive effects. In addition, adjusting the PO_4^{3-} to $\text{P}_2\text{O}_7^{4-}$ ratio in the polyanion framework could modify the covalency and electronic environment around the transition metal centers, potentially affecting both redox behavior and ionic mobility [62].

Ex-situ XRD analysis revealed that ZKFP underwent highly reversible lattice breathing, accommodating internal strain and preserving structural integrity, whereas ZFP exhibited irreversible peak shifts and accumulated residual stress. These findings confirm that $\text{Zn}^{2+}/\text{K}^+$ dual-ion intercalation not only enhances electrochemical kinetics but also plays a crucial role in stabilizing the cathode structure under AZIBs conditions. This work underscores the potential of alkali-ion co-intercalation as an effective strategy to overcome the intrinsic limitations of divalent-ion systems and provides insight into effective cathode design for high-performance AZIBs.

CRediT authorship contribution statement

Sunha Hwang: Writing – original draft, Visualization, Validation, Investigation, Data curation, Conceptualization. **Hyunji Kwoen:** Writing – original draft, Visualization, Formal analysis, Data curation. **Jihoe Lee:** Writing – original draft, Validation, Methodology, Formal analysis, Data curation. **Bonyoung Ku:** Validation, Methodology. **Jinho Ahn:** Methodology, Investigation. **Hoseok Lee:** Validation, Methodology. **Lahyeon Jang:** Visualization, Formal analysis. **Hyeong Jun Kook:** Visualization, Methodology. **Hyungsub Kim:** Data curation, Methodology, Validation. **Dong Ok Shin:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Jongsoo Kim:** Writing – review & editing, Visualization, Supervision, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ensm.2025.104570](https://doi.org/10.1016/j.ensm.2025.104570).

Data availability

Data will be made available on request.

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